

Bus Bunching Control System

Real-Time Bus Tracking, Route Diversion, and Anti-Bunching Dashboard

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Problem Statement

Bus bunching causes multiple buses to arrive together, leading to uneven service and long waiting times for passengers. Manual tracking makes it hard to monitor buses in real time and respond quickly. Current systems also lack proper rerouting to handle traffic or disruptions.

Objectives

Four core goals guide this system's design



Live Map Tracking

Render bus positions in real time on a Leaflet/OpenStreetMap interface.



Bunching Detection

Identify proximity violations and trigger alerts to prevent clustering.



Multi-Route Diversion

Present multiple alternative paths to the same destination during disruptions.



Admin Dashboard

Enable live operator controls — divert, stop, resume, and route selection.

Existing System vs. Proposed System

Existing System

- Passive GPS tracking with no live decision support
- No visual route diversion or alternative path display
- Weak bunching detection and alert mechanisms
- Manual, delayed response to route disruptions

Proposed System

- Real-time WebSocket updates via FastAPI backend
- React + Leaflet map with live bus positions and route lines
- Admin-driven diversion with multiple alternative routes
- Automated proximity alerts and live activity logging

System Modules



Backend Simulation

Generates live bus location and speed data for the Tumkur–Bangalore route.



Frontend Dashboard

React UI with map, bus list, alerts, and admin control panel.



WebSocket Communication

Persistent low-latency channel for pushing live updates to the frontend.



Route Generation & Diversion

Computes and renders alternative paths when a bus is diverted by admin.

Input & Output

System Inputs

- Bus GPS coordinates and speed from backend simulation
- Route geometry: waypoints, stops, and destination points
- Admin commands: **divert, stop, resume, select route**
- Live simulation ticks streamed over WebSocket

System Outputs

- Live bus markers and route polylines on interactive map
- Alternative diversion paths highlighted in distinct colors
- Proximity alerts and real-time bus status indicators
- Activity log and dashboard state updates

Mathematical Formulae — Anti-Bunching Logic

Understanding the core formulas and their application is crucial for effective bus bunching prevention.

Core Formulas

Target Headway = Total Route Distance / Number of Buses

Bunching Threshold = Target Headway $\times$ 0.30

Recovery Threshold = Target Headway $\times$ 0.60

Holding Threshold = Target Headway $\times$ 0.40

These formulas define the optimal spacing and critical thresholds for maintaining efficient bus operations.

¹ Example Calculation

For a **20 km** route with **3 buses**:

Target Headway = $20 / 3 = 6.67$ km

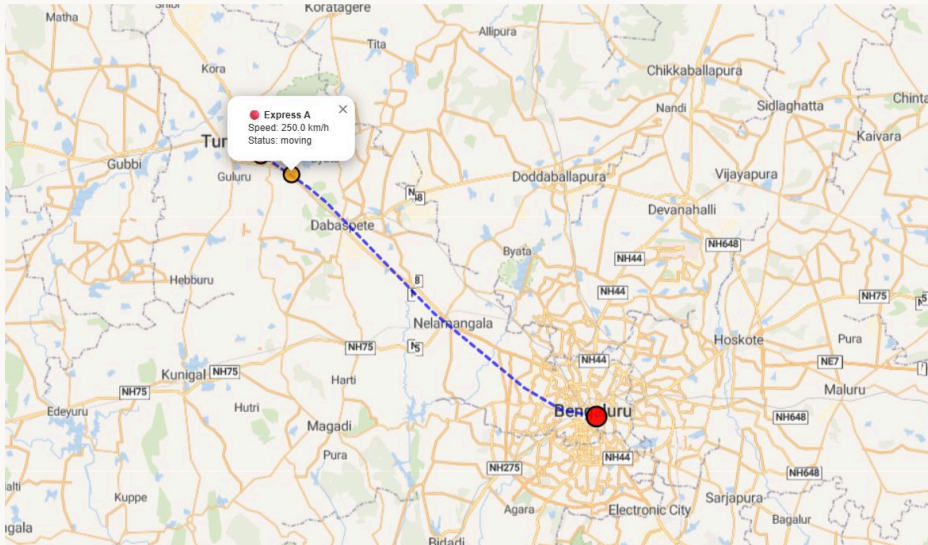
Bunching Threshold = $6.67 \times 0.30 = 2.00$ km

Recovery Threshold = $6.67 \times 0.60 = 4.00$ km

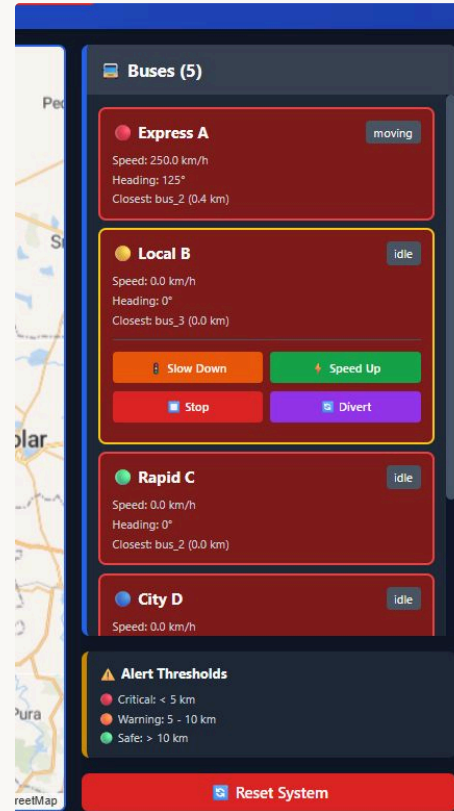
Holding Threshold = $6.67 \times 0.40 = 2.67$ km

This example illustrates how thresholds are derived, guiding real-time decisions to prevent bunching.

Output Dashboard



Sample output -01



Sample output -02

Dashboard Features

- Live map with bus markers, route lines, and stop overlays
- Connection status indicator (connected / disconnected)
- Bus list panel with speed, status, and route info
- Route selection modal showing all diversion alternatives
- Diverted paths highlighted in a distinct accent color
- Real-time alert feed and activity log

Future Work

1

ML-Based Bunching Prediction

Train models to forecast bunching events before they occur using historical and live data.

2

Real GPS Hardware Integration

Replace simulation with physical GPS devices installed on Tumkur-Bangalore buses.

3

Passenger-Facing Mobile App

Extend the system with a public app showing live arrivals and service alerts.

4

Multi-Route & City Scaling

Generalize the platform to support multiple routes and expand to city-wide deployment.

5

Alert & Monitoring

Detects bunching events and logs all system activity in real time.

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